5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

CODE:

```
clc; clear; close all;

%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)
%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');
%% --- Frequency Response Plots ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

OUTPUT:

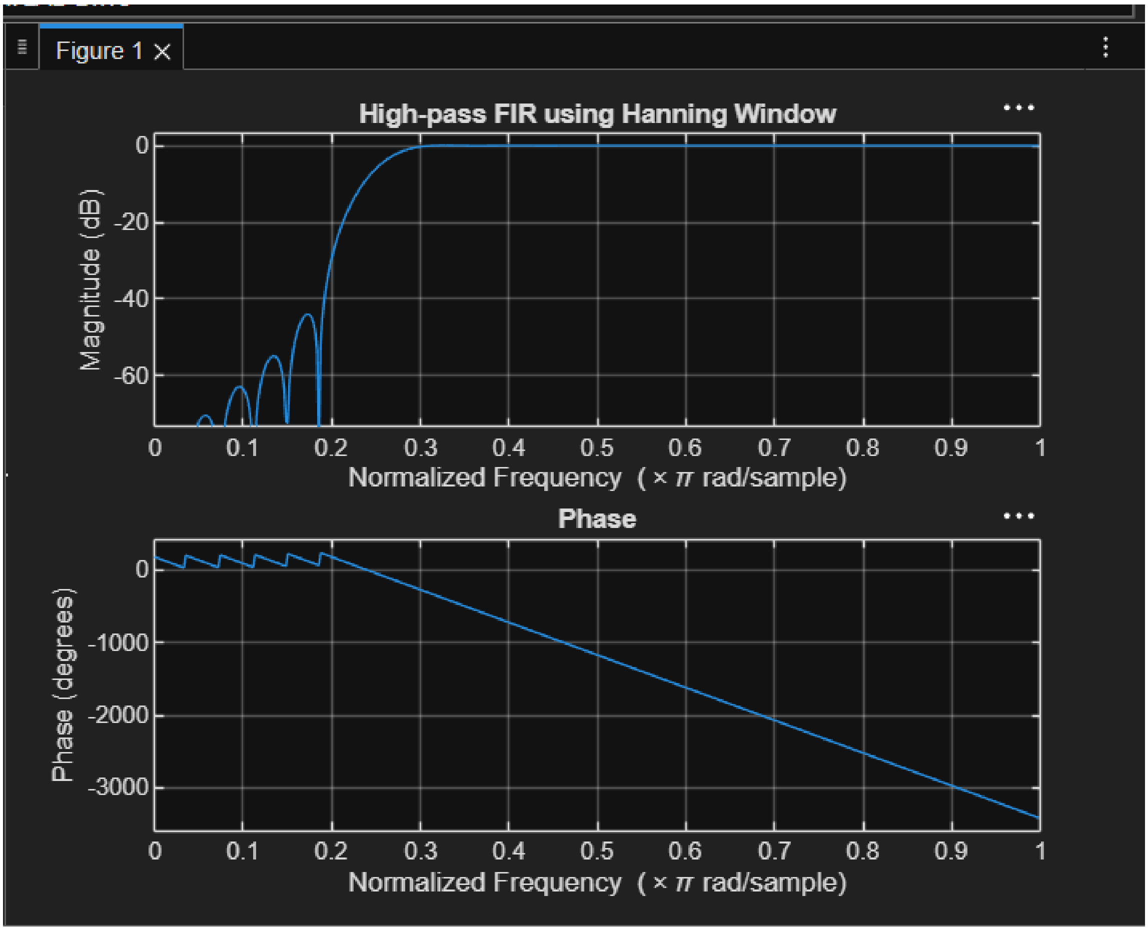
Figure 1
High-pass FIR using Hanning Window
Magnitude (dB)
0
-20
-40
-60
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)